

Statistical Quality Control

Problem-1

A leather tannery manager noticed a high number of complaints about defects in finished leather. During one month, the quality control department recorded 250 defect reports. The defects were classified as follows: 35 cases of color variation, 55 cases of poor softness, 70 cases of uneven dyeing, 25 cases of loose grain, 45 cases of grain damage, and 20 cases of wrinkle marks. To identify the most common defects and prioritize improvement efforts, the management decided to use Pareto analysis. Apply Pareto analysis to the above data and identify which defects should be given the highest priority for improvement.

Solution: Collect data----- Create a data table----- Arrange the data in descending order-----Sum the totals-----Calculate individual percentages-----Calculate cumulative percentages

Defect Type	Frequency	Individual Percentages	Cumulative Percentages
Uneven Dyeing	70	28	28
Poor Softness	55	22	50
Grain Damage	45	18	68
Color Variation	35	14	82
Loose Grain	25	10	92
Wrinkle Marks	20	8	100
	Total= 250	Total= 100	

Uneven Dyeing, Poor Softness, Grain Damage and Color Variation-These four problems occur around 80% problems. So, these defects should be given the highest priority.

Now, Plot the data.

Problem-2

A footwear manufacturing company noticed a high number of customer complaints about defects in finished shoes. During one month, the quality control department recorded 300 defect reports. The defects were classified as follows: 60 cases of sole detachment, 45 cases of stitching defects, 75 cases of color variation, 30 cases of improper fitting, 50 cases of surface scratches, and 40 cases of wrinkles/creasing. Apply Pareto analysis to the above data and identify which defects should be given the highest priority for improvement.

Solution:

Defect Type	Frequency	Individual Percentages	Cumulative Percentages
Color Variation	75	25	25
Sole Detachment	60	20	45
Scratches	50	16.67	61.67
Stitch	45	15	76.67
Wrinkle	40	13.33	90
Improper Fitting	30	10	100
	Total= 300		

Problem-3

Plot the control limits for X bar and R charts for the given sample depth (mm) data.

Subgroup	X ₁	X ₂	X ₃	X ₄	X ₅	X ₆
1	36	39	35	37	38	36
2	41	42	40	43	41	44
3	65	63	55	66	64	62
4	50	48	52	49	51	50
5	28	36	39	37	25	38
6	45	44	46	47	45	41
7	31	33	30	32	35	31

Factors for Computing Central Lines and 3 σ Control Limits for Variables Charts

CHART FOR AVERAGES			CHART FOR STANDARD DEVIATIONS		
Sample Size	Factors for Control Limits		Factor for Central Line	Factors for Control Limits	
n	A ₂	A ₃	C ₄	B ₃	B ₂
2	1.880	2.659	0.7979	0	3.267
3	1.023	1.954	0.8862	0	2.568
4	0.729	1.628	0.9213	0	2.266
5	0.577	1.427	0.9400	0	2.089
6	0.483	1.287	0.9515	0.030	1.970
7	0.419	1.182	0.9594	0.118	1.882
8	0.373	1.099	0.9650	0.185	1.815

CHART FOR RANGES					
Sample Size	Factor for Central Line	Factors for Control Limits			
n	d ₂	D ₃	D ₄	D ₅	D ₆
2	1.128	0	3.267	0	3.865
3	1.693	0	2.574	0	2.745
4	2.059	0	2.282	0	2.375
5	2.326	0	2.114	0	2.179
6	2.534	0	2.004	0	2.055
7	2.704	0.076	1.924	0.078	1.967
8	2.847	0.136	1.864	0.139	1.901

Here, n=6

Solution:

Subgroup	X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	$\bar{X}$	$\bar{R}$
1	36	39	35	37	38	36	36.83	4
2	41	42	40	43	41	44	41.83	4
3	65	63	55	66	64	62	62.5	11
4	50	48	52	49	51	50	50	4
5	28	36	39	37	25	38	33.83	14
6	45	44	46	47	45	41	44.67	6
7	31	33	30	32	35	31	32	5
							$\bar{\bar{X}} = 43.09$	$\bar{\bar{R}} = 6.86$

$\bar{\bar{X}}$ Control Chart

$$\text{Central Limit (CL)} = \bar{\bar{X}}$$

$$\text{Upper Control Limit (UCL)} = \bar{\bar{X}} + A_2 * \bar{R}$$

$$\text{Lower Control Limit (LCL)} = \bar{\bar{X}} - A_2 * \bar{R}$$

For, $\bar{x}$ Control Chart,

$$CL = 43.09$$

$$UCL = 43.09 + 0.483 \times 6.86 = 46.40$$

$$LCL = 43.09 - 0.483 \times 6.86 = 39.78$$

Now, Plot the data

R Control Chart

$$\text{Central Line (CL)} = \bar{R}$$

$$\text{Upper Control Limit (UCL)} = D4 * \bar{R}$$

$$\text{Lower Control Limit (LCL)} = D3 * \bar{R}$$

For $\bar{R}$ Control Chart,

$$CL = 6.86$$

$$UCL = 2.004 \times 6.86 = 13.75$$

$$LCL = 0 \times 6.86 = 0$$

Now, Plot the data

Problem-4

Subgroup	X1	X2	X3	X4	X5	X6
1	34	37	35	36	38	35
2	42	40	43	41	44	42
3	58	61	60	57	62	59
4	49	51	48	52	50	49
5	32	35	34	36	33	35
6	44	46	45	43	47	45
7	38	40	37	39	41	38
8	52	54	51	53	55	52

Plot the control limits for $\bar{X}$ bar and R charts for the given sample.

Problem-5

A footwear manufacturing company produces shoe soles with a specified thickness of 10.00 ± 0.20 mm. The production process is operating under controlled conditions, with no assignable causes of variation. The standard deviation of the process has been estimated to be 0.05 mm.

- (i) What is the value of C_p for this process? What does this value indicate about the capability of the footwear manufacturing process?

- (ii) If the standard deviation of the footwear manufacturing process increases to 5%, how would this change affect the process capability (Cp)?

Solution:

- (i) Given, Target thickness=10.00 mm

$$USL = 10 + 0.2 = 10.2 \text{ mm}$$

$$LSL = 10 - 0.2 = 9.8 \text{ mm}$$

Standard deviation, $\sigma = 0.05 \text{ mm}$,

Process capability index, $C_p = \frac{USL - LSL}{6\sigma} = (10.2 - 9.8) / (6 \times 0.05) = 1.33$; the process is capable of consistently producing parts within the specification limits by meeting accepted standard value

- (ii) Standard deviation, $\sigma = 0.05 + 0.05 \times 5\% = 0.0525 \text{ mm}$

$$\text{New Process capability index, } C_p = \frac{USL - LSL}{6\sigma} = (10.2 - 9.8) / (6 \times 0.0525) = 1.27$$

Problem-6

Subgroup Number	X ₁	X ₂	X ₃	X ₄
1	35	40	32	37
2	46	37	36	41
3	34	40	34	36
4	69	64	68	59
5	38	34	44	40
6	41	40	29	34
7	38	44	28	58
8	35	41	37	38
9	56	55	45	48
10	38	40	45	37
11	39	42	35	40
12	42	39	39	36
13	43	36	35	38

14	39	38	43	44
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Compute the C_p and C_{pk} from the following data table

Solution:

Subgroup Number	X1	X2	X3	X4	Average ($\bar{X}$)	Range (R)
1	35	40	32	37	36	8
2	46	37	36	41	40	10
3	34	40	34	36	36	6
4	69	64	68	59	65	10
5	38	34	44	40	39	10
6	41	40	29	34	36	12
7	38	44	28	58	42	30
8	35	41	37	38	37.75	6
9	56	55	45	48	51	11
10	38	40	45	37	40	8
11	39	42	35	40	39	7
12	42	39	39	36	39	6
13	43	36	35	38	38	8
14	39	38	43	44	41	6
Xbar					41.41071429	
R bar					9.857142857	

Xbar			R bar		
UCL	CL	LCL	UCL	CL	LCL
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0
48.6	41.41	34.22	22.49	9.857	0

For calculating C_p and C_{pk} ,

We know,

Process Capability Ratio (PCR) = $C_p = (USL - LSL) / 6\sigma$

$$C_{pk} = \frac{\text{Min} \{ (USL - \bar{X}) \text{ or } (\bar{X} - LSL) \}}{3\sigma}$$

$$\sigma = \bar{R}/d_2$$

Process Capability Ratio is relevant to X bar control chart,

So the upper control limit (UCL) and lower control limit (LCL) from X bar control chart will be applicable for calculating C_p and C_{pk}

So, $\sigma = 9.857/2.059 = 4.787$ (For $n=4$, d_2 is equal to 2.059)

Process Capability Ratio (PCR) = $C_p = (USL - LSL) / 6\sigma$

Sample Size	CHART FOR RANGES					Chart for Medians
	Factor for Central Line	Factors for Control Limits				
		d_2	D_3	D_4	D_5	
2	1.128	0	3.267	0	3.865	2.224
3	1.693	0	2.574	0	2.745	1.265
4	2.059	0	2.282	0	2.375	0.829
5	2.326	0	2.114	0	2.179	0.712
6	2.534	0	2.004	0	2.055	0.562
7	2.704	0.076	1.924	0.078	1.967	0.520
8	2.847	0.136	1.864	0.139	1.901	0.441

$$= (48.6 - 34.22) / (6 \times 4.787) = 0.5$$

$$C_{pk} = \frac{\text{Min} \left\{ (USL - \bar{X}) \text{ or } (\bar{X} - LSL) \right\}}{3\sigma}$$

Here, $USL - \bar{X} = 48.6 - 41.41 = 7.19$

$\bar{X} - LSL = 41.41 - 34.22 = 7.19$

$$C_{pk} = 7.19 / (3 \times 4.787) = 0.5$$

Problem-7

Alexon-a footwear manufacturing company uses an acceptance sampling plan to inspect batches of finished shoes before shipment. A lot contains 1,200 pairs of shoes, and inspection reveals 36 defective pairs. Based on the sampling plan, the probability of accepting the lot (P_a) is 0.75. Calculate the Average Outgoing Quality (AOQ) of the accepted lots and interpret the result.

Solution:

- Lot size (N)=1200
- Actual defectives in the lot (d)=36
- $P_a = 0.75$
- So, $P = \frac{d}{N} = 36/1200 = 0.03$

AOQ = $P_a \times P = 0.75 \times 0.03 = 0.0225$; 2.25% defective items in outgoing lots.

TQM Tools and Techniques-ii

A footwear manufacturer produces finished shoe soles with a specified thickness. According to the company's quality specification system:

- Lower Specification Limit (LSL) = 8.5 mm
- Target thickness = 9.0 mm
- Upper Specification Limit (USL) = 9.5 mm

The company considers any sole thickness between 8.5 and 9.5 mm to be acceptable. However, the production manager has observed that soles with thicknesses close to either specification limit may cause problems during sole attachment, shoe balance, and durability, even though they technically meet the specification. The company estimates that when the sole thickness reaches either specification limit, the resulting quality-related cost to the company/customer is Tk 150 per pair.

During inspection, five sole samples are found with the following thicknesses:

- Sample A = 9.0 mm
- Sample B = 9.2 mm
- Sample C = 9.4 mm
- Sample D = 8.7 mm
- Sample E = 9.5 mm

Questions:

- a. Identify the target value, tolerance, and maximum loss
- b. Calculate the quality loss for each sample.
- c. Which piece has the highest quality loss?